

JEE MAIN REVISION MASTER: THREE DIMENSIONAL GEOMETRY 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 41 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Three Dimensional Geometry - Core Application	41 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Three Dimensional Geometry - Core Application (Very High Yield — 41 PYQs)

Core Principle: Key problem pattern in Kcet Three Dimensional Geometry.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B=90^\circ$, then the value of "a" is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "The measure of the angle between the lines $x=k+1, y=2k-1, z=2k+3, k \in \mathbb{R}$ and $(x-1)/2=(y-2)/1=(z-3)/1$ is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "The angle between the lines whose direction ratios are a,b,c and $b-c, c-a, a-b$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The three points $A(2,4,3)$, $B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is

- (A) 1 or 4
- (B) -1 or 4 **[Correct]**
- (C) 1 or -4
- (D) -1 or -4

Correct Answer: B

Solution Analysis: Since the triangle is right-angled at B, the two sides meeting at B must be perpendicular. So, vectors $\vec{BA}$ and $\vec{BC}$ must satisfy $\vec{BA} \cdot \vec{BC} = 0$. Now, $A(2,4,3), B(4,a,9), C(10,-1,7)$. So, $\vec{BA} = A - B$
 $\vec{BA} = (2-4, 4-a, 3-9) = (-2, 4-a, -6)$ Also, $\vec{BC} = C - B$
 $\vec{BC} = (10-4, -1-a, 7-9) = (6, -1-a, -2)$ Since they are perpendicular, $(-2)(6) + (4-a)(-1-a) + (-6)(-2) = 0$
 $-12 + (4-a)(-1-a) + 12 = 0$
 $(4-a)(-1-a) = 0$ So, $4-a=0$ or $-1-a=0$ Hence, $a=4$ or $a=-1$ So the correct option is Option B-1 or 4